Regime-mixture exceedance probabilities from a partly observed path, with collapse diagnostics

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Abstract

Many tasks ask for the probability that a process ends above a threshold, given part of its path, while its spread switches between activity levels. We release `regimix`, which fits a label-free hidden Markov model (unobserved switching states) to unit-time scaled lookback segments, mixes per-regime Gaussian tail probabilities by the filtered regime weights, and calibrates the mixture out of fold. On EUR/USD 1-minute episodes we find that regimix ($K = 2$) beats the one-regime pooled probit by 0.0079 nats of log-loss (paired moving-block bootstrap SE 0.0014), but ties a realised-volatility probit (+0.0024, SE 0.0015) and loses to a HAR-style probit (+0.0043, SE 0.0012). Across 3 public datasets it ties or loses to the best volatility probit, and appending its outputs to gradient boosting changes log-loss by +0.0004 / +0.0011 / +0.0015 (SE 0.0004 / 0.0008 / 0.0004), never better. All numbers come from one held-out tail of time per dataset and 3 seeds, with bootstrap SEs over time blocks or series.

1 Problem

An episode i is a scalar process X observed up to a decision time t_i, a threshold k_i and a remaining horizon $\tau_i > 0$. The label is

$$y_i = \mathbf{1}\{X(t_i + \tau_i) > k_i\}, \tag{1}$$

with $\mathbf{1}\{\cdot\}$ the indicator of the event, and the goal is a calibrated $P(y_i = 1)$ from the path before t_i. The plain answer is a probit, a Gaussian tail probability whose width comes from the recent path, which works when volatility is stable but fails when activity jumps between a quiet and a busy level faster than a long realised-volatility window (the summed squared recent increments) can follow.

We expected a filtered regime, read off a few lookback windows, to react within one window and beat a slow volatility estimate. The filter is Hamilton's [Hamilton, 1989] and the regime-weighted normal tail is standard in regime-switching return models [Hardy, 2001]; we claim neither. The delta is a label-free fit over many short episodes instead of one long series, a cross-fitted calibration, a variance-floor fix to the upstream library [hmmlearn developers, 2024], and diagnostics that say when the regimes are not real. The result is mostly negative, and we report it as such.

Preprint.

2 The change

Pick lookback edges $0 = e_0 < e_1 < \cdots < e_S$. Segment s is the increment over one window, scaled to unit time:

$$z_{i,s} = \mathrm{clip}\left(\frac{X(t_i - e_{s-1}) - X(t_i - e_s)}{\sqrt{e_s - e_{s-1}}},\ -c,\ c\right), \tag{2}$$

$$w_i = \alpha_{i,S}\, A, \qquad \alpha_{i,S}(j) = P(r_S = j \mid z_{i,1}, \ldots, z_{i,S}), \tag{3}$$

$$\tilde{p}_i = \sum_{j=1}^{K} w_i(j)\, \Phi\left(\frac{d_i + \mu_j \tau_i}{\max(\sigma_j, \sigma_{\min})\sqrt{\tau_i}}\right), \qquad d_i = X(t_i) - k_i, \tag{4}$$

$$p_i = \mathrm{sigmoid}\big(a + b\ \mathrm{logit}\,\tilde{p}_i\big). \tag{5}$$

Symbol	Meaning	Symbol	Meaning
$z_{i,s}$	segment s, oldest first	c	optional clip
r_s, K	regime at segment s; regime count	A	transition matrix
$\alpha_{i,S}$	filtered regime probabilities	w_i	horizon regime weights
μ_j, σ_j^2	drift, variance per unit time ($\mu_j = 0$)	$\sigma_{\min}$	spread floor (0.05)
Φ	normal CDF	a, b	Platt parameters [Platt, 1999, Lin et al., 2007]

We fit the hidden Markov model (HMM) with K Gaussian states by Baum–Welch [Baum et al., 1970] on the segment sequences of all episodes, without labels. Equation (3) is its forward pass, batched over episodes, and Equation (4) averages one random-walk tail probability per regime. Labels enter only in Equation (5), through two parameters that we fit on outputs from 5 contiguous time-block folds before the HMM is refitted on all training rows. The estimator also returns w_i and $\mathrm{logit}\,\tilde{p}_i$, so that a downstream learner can use the regimes as inputs. We claim three properties, and Appendix A proves each of them.

Lemma 1 (exact mixture)**.** *If, given the horizon regime $r = j$, $X(t_i + \tau_i) - X(t_i) \sim \mathcal{N}(\mu_j \tau_i, \sigma_j^2 \tau_i)$ independently of $z_{i,1..S}$, $P(r = j \mid z_i) = w_i(j)$ and $\sigma_j \geq \sigma_{\min}$, then $\tilde{p}_i = P(y_i = 1 \mid z_i)$.*

Lemma 2 (no test label enters)**.** *If no training label reads a value of X inside the held-out period, then p_i for a held-out episode depends on labels only through training labels.*

The second part of the change is a variance floor: hmmlearn documents `min_covar` as a floor on the covariance but applies it only at initialisation, so on data with many exact zeros one state collapses onto a spike at zero. Its likelihood then grows without bound and its posterior turns into a one-bit gate. Our `FlooredGaussianHMM` applies the floor after every M-step (the parameter-update half of EM):

$$v_j \leftarrow \max\big(v_j^{\mathrm{EM}},\ m\big)\,. \tag{6}$$

Here v_j^{EM} is the unconstrained variance update and m is `min_covar` (0.05 by default).

Lemma 3 (monotone EM)**.** *For diagonal or spherical covariances, update (6) is the exact maximiser of the M-step objective over $\{v \geq m\}$, so the log-likelihood never decreases.*

We turn the conditions under which regimes can help into warnings, computed from the fitted variances v_j and the self-transition probabilities A_{jj}:

$$\rho = \frac{\min_j v_j}{m}, \qquad D_j = \frac{1}{1 - A_{jj}}, \qquad R = \frac{\max_j \sigma_j}{\min_j \sigma_j}. \tag{7}$$

Here ρ is the collapse ratio (near 1: a state sits on the floor), D_j the mean dwell in segments and R the spread ratio. A dwell below 2 segments or a spread ratio below 2 triggers a warning; the report also gives the share of exactly flat segments, the count of distinct posterior values (a gate has two) and per-segment permutation importance.

3 Setup

Data. We use 3 public datasets, each chosen because its activity level switches. *EUR/USD 1-minute* bars from HistData, 2016–2023: at a random minute each hour, $P(\text{price in 30 min} > \text{price} + u)$ with

$u \in \{-3, 0, +3\}$ pips. *US large-cap daily* closes for 59 tickers, 2005–2024, one decision day in 4 per ticker: $P(\text{close in 10 days} > \text{close} \times (1 + u))$, $u \in \{-2, 0, +2\}$ %. *M4 hourly* [Makridakis et al., 2020], one decision in 9 hours: $P(\log y_{t+h} > \log y_t + u)$ for $h \in \{12, 24\}$ hours and $u = c\,\hat{\sigma}_1\sqrt{h}$, $c \in \{-0.5, 0, 0.5\}$. Appendix B lists the lookback edges. The held-out set is the last 30 % of time (per series for M4) and is never shuffled; we drop every training row whose label reads into it, an embargo (a gap against look-ahead) of one horizon. That leaves 34,202 / 13,078, 51,360 / 22,125 and 51,952 / 23,288 training / held-out episodes, each series scaled by the spread of its one-step training increments.

Arms. All arms see the same segments, distances and horizons, and we give each a Platt map fitted on training rows. *regimix* runs its defaults ($K = 2$, `n_iter` 60, `tol` 0.01, floor and next step on, drift off), and the *one-regime pooled probit* is the same model with $K = 1$. The *realised-vol probit* uses the realised variance per unit time over the lookback; the *HAR-style probit* (heterogeneous autoregressive volatility) regresses log realised variance over the horizon on its value at each lookback window [Corsi, 2009, Andersen et al., 2003]. The *gate ablation* replaces w_i by a one-hot "last segment large". *Logistic regression* and *histogram gradient boosting* [Pedregosa et al., 2011] get $(Z, |Z|, d, \tau, d/\sqrt{\tau})$. *Stock EM* is the pristine upstream M-step [hmmlearn developers, 2024] without the floor, and the stacking test [Wolpert, 1992] appends out-of-fold w_i and $\operatorname{logit} \tilde{p}_i$ to the boosting inputs.

Metric and error. Log-loss in nats, averaged over 3 seeds; the paired difference per episode is

$$\Delta_i = \ell(p_i^{\text{regimix}}, y_i) - \ell(p_i^{\text{arm}}, y_i), \qquad \ell(p, y) = -y \log p - (1 - y) \log(1 - p). \tag{8}$$

Negative $\bar{\Delta}$ means regimix is better. Episodes in one time block share a path, so a per-episode SE is too small (FX against $K = 1$: 0.0011 against 0.0014). We therefore sum Δ_i within blocks and run a moving-block bootstrap [Künsch, 1989] (resampling runs of consecutive blocks) with 2,000 draws over 749 FX days and 1,500 equity dates, with the block length set by the first lag at which the autocorrelation of block means falls inside $2/\sqrt{n}$. M4 series are independent, so there we resample 414 whole series (a Newey–West SE [Newey and West, 1987] is kept as a check). We call a difference a tie when its 95 % interval covers zero. Calibration is the expected calibration error over 15 bins (ECE, the mean gap between predicted and observed frequency) [Guo et al., 2017]. The 15 arms of the main run were written down before any held-out number existed. After the main run lost to the volatility probits we tried 6 post-hoc variants on the same tail, about 18 comparisons, and kept a variant only if it won on all 3 datasets.

4 Results

No dataset gives regimix a win over the best volatility probit (Table 1). On EUR/USD the regimes we find are real: they persist 19.6 and 8.7 segments on average and their spreads differ 3.37-fold. There regimix beats the pooled probit by 0.0079 nats and the gate ablation by 0.0075. The realised-vol probit is a tie (+0.0024, interval -0.0005 to +0.0053) and the HAR probit wins by 0.0043 (+0.0019 to +0.0067), because a two-level regime only coarsens a level that the HAR arm reads directly.

Table 1: Regimix wins against the pooled probit only where regimes exist, and never against the best volatility probit. Held-out log-loss in nats; in brackets regimix minus arm with its block or cluster bootstrap SE (negative: regimix better). The last row is boosting with regime outputs minus boosting alone (positive: appending hurt).

arm	EUR/USD 1-min	US large-cap daily	M4 hourly
regimix ($K = 2$)	**0.6201**	**0.6505**	**0.5878**
pooled probit ($K = 1$)	0.6280 (-0.0079 ± 0.0014)	0.6508 (-0.0003 ± 0.0005)	0.5849 (+0.0029 ± 0.0002)
realised-vol probit	0.6177 (+0.0024 ± 0.0015)	0.6505 (0.0000 ± 0.0008)	0.5869 (+0.0010 ± 0.0006)
HAR-style probit	0.6158 (+0.0043 ± 0.0012)	0.6496 (+0.0008 ± 0.0005)	0.5885 (-0.0007 ± 0.0003)
gate ablation	0.6276 (-0.0075 ± 0.0012)	0.6541 (-0.0036 ± 0.0005)	0.5910 (-0.0032 ± 0.0003)
logistic regression	0.6288 (-0.0087 ± 0.0016)	0.6523 (-0.0018 ± 0.0008)	0.3476 (+0.2402 ± 0.0049)
gradient boosting	0.6195 (+0.0006 ± 0.0010)	0.6530 (-0.0025 ± 0.0008)	0.2126 (+0.3752 ± 0.0053)
boosting + regime − boosting	+0.0004 ± 0.0004	+0.0011 ± 0.0008	+0.0015 ± 0.0004

On equities (Figure 1) the regimes pass both warnings (dwell 14.9 / 3.9, spread ratio 2.72), yet every probit lands within 0.0008 nats of regimix. On M4 the diagnostics warn on both counts (dwell 1.0 /

1.3 segments, spread ratio 1.09), and M4 is exactly where regimix loses to $K = 1$, by 0.0029 nats. We count its edge over the HAR probit there (-0.0007, interval -0.0013 to 0.0000) as a tie. On M4 logistic regression and boosting reach 0.3476 and 0.2126, far below every probit, because a driftless random walk cannot express its daily seasonality and mean reversion.

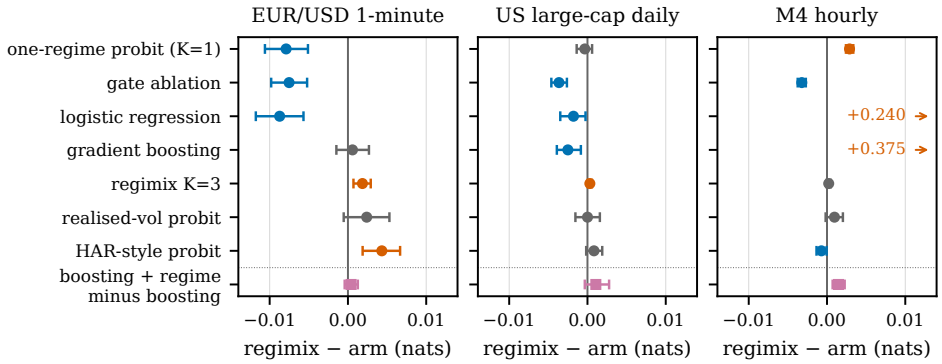


Figure 1: The volatility probits match or beat regimix on every dataset, and the regime outputs never help boosting. Paired held-out log-loss difference, regimix minus arm, with 95 % block (EUR/USD, equities) or cluster (M4) bootstrap intervals; blue is a regimix win, red a loss, grey a tie, purple the stacking row. Off-scale values printed.

Appended to boosting, the regime outputs change log-loss by +0.0004 and +0.0011 (ties) and by +0.0015 on M4, where the interval +0.0007 to +0.0023 excludes zero on the wrong side.

Calibration follows the same order (Figure 2): on EUR/USD the seed-averaged ECE is 0.0181 for regimix, 0.0237 for the pooled probit and 0.0122 for HAR, while on M4 regimix is the worst calibrated of the three at 0.0623.

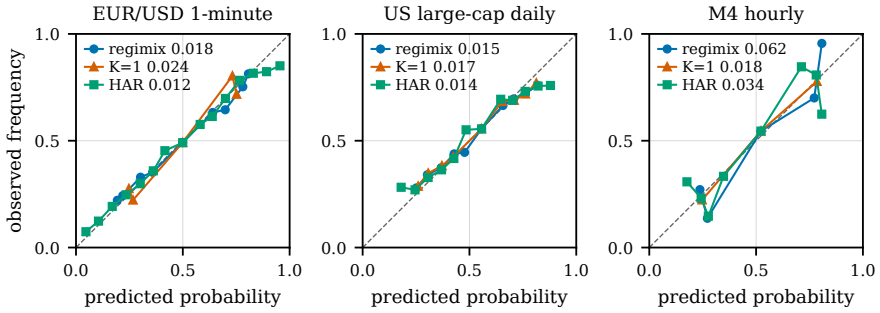


Figure 2: Regimix is better calibrated than the pooled probit where regimes exist and worse on M4, where they do not. Reliability diagrams of the seed-averaged held-out probabilities, 15 bins, sparse bins hidden; the legend gives each arm's ECE.

Of the single-default ablations (Table 2, Appendix B) only $K = 3$ helps (+0.0019 on EUR/USD, +0.0003 on equities), still short of the volatility probits; we keep $K = 2$ because $K = 3$ was picked on the held-out tail, and no post-hoc variant won on all 3 datasets.

5 Limits

The idea lost a comparison to a continuous volatility probit and stays open: that baseline is label-light and closed-form, so it meets every design constraint we set, and a gain needs what the level misses, such as switching inside the horizon [Hardy, 2001], which we did not test. The evidence is thin: one held-out tail per dataset, 3 seeds, post-hoc variants on that same tail, a floor fix with only synthetic evidence, survivorship bias in the equities, and M4 rows that speak only about probits. Appendix B adds a drift unit mismatch and our one off-benchmark result, also no win.

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A Proofs

Lemma 1. **Given:** 1) law of total probability over a partition $\{r = j\}_{j=1}^{K}$; 2) given $r = j$, the increment $U = X(t_i + \tau_i) - X(t_i) \sim \mathcal{N}(\mu_j \tau_i, \sigma_j^2 \tau_i)$, independent of z_i; 3) $P(r = j \mid z_i) = w_i(j)$; 4) $\Phi(-x) = 1 - \Phi(x)$.

Choose: the event $E = \{y_i = 1\} = \{X(t_i) + U > k_i\} = \{U > -d_i\}$ and the partition of rule 1.

Need to prove: $P(E \mid z_i) = \sum_j w_i(j)\, \Phi\big((d_i + \mu_j \tau_i)/(\sigma_j \sqrt{\tau_i})\big)$.

$$
\begin{aligned}
P(E \mid z_i) &= \textstyle\sum_j P(E \mid r = j, z_i)\, P(r = j \mid z_i) && \text{rule 1} \\
&= \textstyle\sum_j P(E \mid r = j)\, w_i(j) && \text{independence in rule 2, rule 3} \\
P(E \mid r = j) &= P\Big(\tfrac{U - \mu_j \tau_i}{\sigma_j \sqrt{\tau_i}} > \tfrac{-d_i - \mu_j \tau_i}{\sigma_j \sqrt{\tau_i}}\Big) && \text{standardise } U, \text{ rule 2} \\
&= 1 - \Phi\Big(\tfrac{-d_i - \mu_j \tau_i}{\sigma_j \sqrt{\tau_i}}\Big) && \text{definition of } \Phi \\
&= \Phi\Big(\tfrac{d_i + \mu_j \tau_i}{\sigma_j \sqrt{\tau_i}}\Big) && \text{rule 4}
\end{aligned}
$$

Precondition: Equation (4) uses $\max(\sigma_j, \sigma_{\min})$, which equals σ_j because $\sigma_j \geq \sigma_{\min}$. **Therefore,** substitution gives $\tilde{p}_i = P(y_i = 1 \mid z_i)$. □

Lemma 3. **Given:** 1) generalised EM [Dempster et al., 1977]: if each update satisfies $Q(\theta^{\text{new}} \mid \theta^{\text{old}}) \geq Q(\theta^{\text{old}} \mid \theta^{\text{old}})$, the log-likelihood does not decrease; 2) for diagonal or spherical covariances Q separates into one term per state and coordinate, and the variance term has the form $q(v) = -\frac{1}{2}(\nu \log v + s/v)$ with weight $\nu > 0$ and scatter $s > 0$ (a conjugate variance prior only changes ν and s); 3) the mean update does not depend on v.

Choose: one variance term q and the feasible set $v \geq m$.

Need to prove: $\arg\max_{v \geq m} q(v) = \max(s/\nu,\, m)$.

$$
\begin{aligned}
q'(v) &= -\tfrac{1}{2}\left(\tfrac{\nu}{v} - \tfrac{s}{v^2}\right) && \text{derivative of rule 2} \\
&= \tfrac{\nu}{2v^2}\left(\tfrac{s}{\nu} - v\right) && \text{factor out } \tfrac{\nu}{2v^2} > 0
\end{aligned}
$$

So $q' > 0$ for $v < s/\nu$ and $q' < 0$ for $v > s/\nu$: q is unimodal with peak $v^* = s/\nu = v^{\text{EM}}$. If $v^* \geq m$ the peak is feasible and is the maximiser. If $v^* < m$ then q is decreasing on $[m, \infty)$ and the maximiser is m. Both cases give $\max(v^*, m)$, which is update (6).

Precondition: rule 1 needs θ^{old} feasible, so that it is a candidate of the constrained maximisation. The initial variances carry $+m$, so they are feasible. Every later variance is the output of (6), so it is $\geq m$. By induction every iterate is feasible.

Therefore, by rule 3 the means keep their usual update, each variance gets its exact constrained maximiser, Q does not decrease, and by rule 1 neither does the log-likelihood. □

Lemma 2. **Given:** 1) $\theta = \text{fit}(Z_{\text{train}})$, Baum–Welch on segments only; 2) $(a, b) = \text{Platt}\big(\{\text{logit}\, \tilde{p}_j^{\text{oof}}, y_j\}_{j \in \text{train}}\big)$, where each out-of-fold output comes from an HMM fitted on segments only; 3) for a held-out episode, $p_i = g(z_i, d_i, \tau_i; \theta, a, b)$ by Equations (3)–(5).

Need to prove: p_i depends on labels only through $\{y_j\}_{j \in \text{train}}$.

$$
\begin{aligned}
p_i &= g(z_i, d_i, \tau_i;\, \theta,\, a,\, b) && \text{rule 3} \\
&= g\big(z_i, d_i, \tau_i;\, \text{fit}(Z_{\text{train}}),\, \text{Platt}(\{\tilde{p}_j^{\text{oof}}, y_j\}_{j \in \text{train}})\big) && \text{rules 1 and 2}
\end{aligned}
$$

The only label symbols on the right are y_j with $j \in$ train.

Precondition: a training label must not itself carry held-out information. A training label reads $X(t_j + \tau_j)$; rows with $t_j + \tau_j$ inside the held-out period are dropped (the embargo of Section 3), so every remaining y_j is a function of X before the held-out period.

Therefore, no held-out label enters p_i. The out-of-fold step is not needed for this; it only makes the calibrator see raw outputs from an HMM that did not fit those rows, as at test time. □

B Ablations and post-hoc variants

The benchmark arm equals the public `RegimeExceedanceClassifier` with a maximum logit difference of 0. The lookback edges are 0, 1, 3, 7, 15, 31, 63 minutes (EUR/USD), 0, 1, 3, 7, 15, 31 days (equities) and 0, 1, 2, 4, 8, 16, 32, 48 hours (M4). The tie on M4 for $K = 3$ has interval -0.0001 to +0.0005. Turning drift on is a large loss on equities and M4; behind `drift=True` the segment mean is drift per $\sqrt{\text{time}}$ while Equation (4) reads it per unit time. Neither floor binds on the public data: the smallest stock-EM variance is 4.8, 8.3 and 30.5 times the floor, so both ablations are bit-identical. The floor's evidence is synthetic. On 4,000 sequences of length 7 drawn as rounded $\mathcal{N}(0, 0.5^2)$ values (about 68 % exact zeros), stock `GaussianHMM` with $m = 10^{-3}$ ends with variances 0.163 and 2.3×10^{-6}, which is 0.002 times the floor.

Table 2: No single default carries the method, and the floors never bind on public data. Regimix minus the ablated arm, log-loss in nats ± paired SE; a positive value means the ablated arm is better.

ablation	EUR/USD 1-min	US large-cap daily	M4 hourly
$K = 3$	+0.0019 ± 0.0006	+0.0003 ± 0.0001	+0.0002 ± 0.0001
stock EM (no floor)	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
next step off	+0.0001 ± 0.0002	-0.0005 ± 0.0001	-0.0003 ± 0.0000
$\sigma_{\min}$ off	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
drift on	0.0000 ± 0.0001	-0.0093 ± 0.0028	-0.1028 ± 0.0024
isotonic calibration	0.0000 ± 0.0004	-0.0003 ± 0.0003	+0.0076 ± 0.0006
per-regime Platt	+0.0002 ± 0.0003	-0.0010 ± 0.0002	+0.0020 ± 0.0003

All 6 variants were tried after the main run, on the same held-out tail. Values are variant minus the named arm, ± paired SE (negative: the variant is better). The level-relative variants divide segments and distance by the realised-vol or HAR level, so the regimes model only the change around it.

Table 3: Dividing by the volatility level turns regimix into a near-copy of the level arm, and more regimes do not help against it. No variant wins on all three datasets.

variant	against	EUR/USD 1-min	US large-cap daily	M4 hourly
$K = 4$	realised vol	-0.0003 ± 0.0012	+0.0001 ± 0.0008	+0.0031 ± 0.0007
$K = 4$	HAR	+0.0016 ± 0.0010	+0.0009 ± 0.0005	+0.0015 ± 0.0006
rv-relative, $K = 2$	realised vol	-0.0002 ± 0.0001	0.0000 ± 0.0000	+0.0022 ± 0.0003
rv-relative, $K = 3$	realised vol	-0.0005 ± 0.0002	0.0000 ± 0.0000	+0.0060 ± 0.0005
HAR-relative, $K = 2$	HAR	0.0000 ± 0.0001	0.0000 ± 0.0000	+0.0022 ± 0.0002
HAR-relative, $K = 3$	HAR	-0.0001 ± 0.0000	0.0000 ± 0.0000	+0.0011 ± 0.0002
$n = 300$	realised vol	+0.0035 ± 0.0015	+0.0051 ± 0.0011	-0.0041 ± 0.0007
$n = 1,000$	realised vol	+0.0021 ± 0.0015	-0.0016 ± 0.0007	-0.0026 ± 0.0006
$n = 3,000$	realised vol	+0.0032 ± 0.0017	-0.0004 ± 0.0008	-0.0028 ± 0.0006
$n = 300$	boosting	-0.0281 ± 0.0023	-0.1556 ± 0.0087	+0.1881 ± 0.0046

The only result we have outside these benchmarks follows. On a proprietary binary threshold-exceedance task, the regime-mixture output did not beat a single-regime realised-volatility probit on its own: block-CV log-loss +0.010 nats worse (paired block-bootstrap SE 0.012, 95 % CI [-0.013, +0.032], i.e. a tie within noise). Against stock hmmlearn on held-out log-loss, a zero-inflated emission gave +0.0001 (SE 0.0037), a per-position variance +0.0014 (SE 0.0012), and a Student-t emission -0.0040 (SE 0.0015, 95 % CI [-0.0068, -0.0009]; negative is better).

The small-n rows train on the most recent 300, 1,000 or 3,000 episodes. With few labels regimix beats boosting by a wide margin on EUR/USD and equities, but not the realised-vol probit. M4 training rows are ordered by series, so its small-n rows test transfer to other series, not a learning curve.